Probability: Introduction

Questions 1–57

Question 1. Yusuf flips two fair coins simultaneously.

a) Draw a sample space diagram to show all of the possible outcomes.

b) Find the probability that there is at least one tails.

Question 2. Sylvia rolls two fair dice, both of which have four faces numbered 1 to 4. The product of the two scores is to be investigated.

a) Draw a sample space diagram to show all of the possible outcomes.

b) Find the probability that Sylvia scores a product greater than 10

Question 3. Joanne has two spinners. The first spinner has four equal sections numbered with the first four prime numbers, and the second spinner has three equal sections numbered with the first three square numbers. Joanne spins both spinners simultaneously and adds the scores together.

a) Draw a sample space diagram to show all of the possible outcomes.

b) Find the probability that Joan's score is an even number.

Question 4. Matt has two packs of cards. Each pack contains three cards, one labelled A, one labelled B and one labelled C. Matt randomly selects a card from the first pack, and then randomly selects a card from the second pack.

a) Draw a sample space diagram to show all of the possible outcomes.

b) Find the probability that Matt selects two cards that are both consonants.

Question 5. Kenneth has a covered box that contains 3 red balls, 2 green balls and 1 blue ball. He draws two balls from the box, one after the other, without replacement.

a) Write down a list of all the possible outcomes.
b) Find the probability that Kenneth does not draw the blue ball.

Question 6. Halima has two bags of marbles. The first bag has 3 red marbles and 3 blue marbles. The second bag has 4 red marbles and 4 blue marbles. She draws a marble from each of the bags without replacement.

a) Write down a list all the possible outcomes.
b) Find the probability that Janie draws two red marbles.
c) Find the probability that the two marbles are different colours.
d) Find the probability that one bag now has more red marbles than blue marbles, and the other bag now has more blue marbles than red marbles.

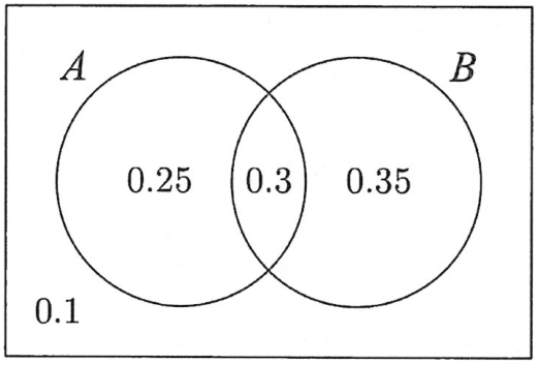
Question 7. A lottery is played in a town hall. There are two buckets, both of which contain balls numbered 0, 1, 2, 3, 4 and 5. A ball is randomly selected from the first bucket and this becomes the first digit of the winning lottery number. A ball is then randomly selected from the second bucket and this becomes the second digit of the winning lottery number.

a) Draw a sample space diagram to show all of the possible outcomes.
b) Find the probability that the winning lottery number is a nonzero multiple of 11
c) Find the probability that the winning lottery number is a nonzero multiple of 4

Question 8. Victor has a bag of marbles. The bag contains 2 red marbles, 2 blue marbles, and 1 green marble. Victor draws three marbles from the bag without replacement.

a) Write down a list of all the possible outcomes.
b) Find the probability that Victor draws two red marbles.
c) Find the probability that Victor draws marbles of all three colours.
d) Find the probability that the bag now only contains marbles of one colour.

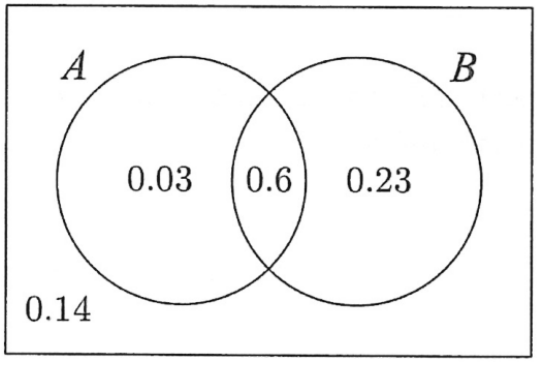
Question 9. Events A and B are shown in the Venn diagram below.



Find

a) $\mathrm{P}(A)$
b) $\mathrm{P}(A \cap B)$
c) $\mathrm{P}(B')$
d) $\mathrm{P}(A \cup B)$
e) $\mathrm{P}(A \cap B')$
f) $\mathrm{P}(A \cup B')$

Question 10. Events A and B are shown in the Venn diagram below.



Find

a) $\mathrm{P}(B)$
b) $\mathrm{P}(A' \cap B)$
c) $\mathrm{P}(A' \cup B)$
d) $\mathrm{P}((A \cup B)')$
e) $\mathrm{P}((A \cap B)')$
f) $\mathrm{P}(A' \cup B')$

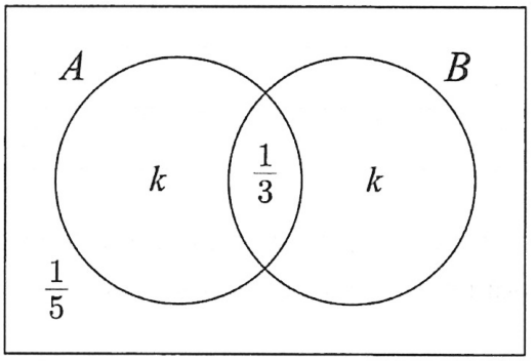
Question 11. Draw a Venn diagram for events A and B, given that $\mathrm{P}(A \cap B) = 0.2$, $\mathrm{P}(A) = 0.3$ and $\mathrm{P}(B) = 0.4$

Question 12. Draw a Venn diagram for events C and D, given that $\mathrm{P}(C \cap D) = 0.7$, $\mathrm{P}(C) = 0.75$ and $\mathrm{P}(D) = 0.85$

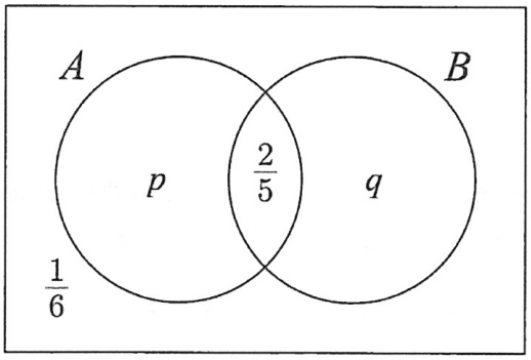
Question 13. Events A and B are shown in the Venn diagram below.

a) Find the value of k

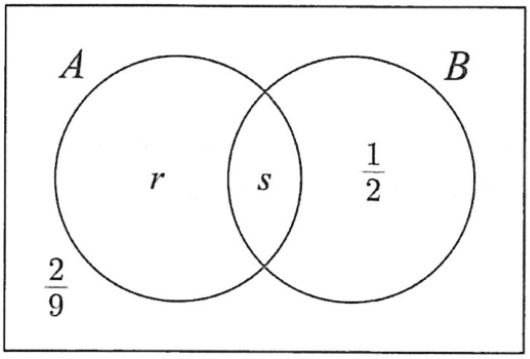
b) Find $\mathrm{P}(A)$



Question 14. Events A and B are shown in the Venn diagram below. It is given that $\mathrm{P}(A) = 2\mathrm{P}(B)$. Find the value of p and the value of q



Question 15. Events A and B are shown in the Venn diagram below. It is given that $2\mathrm{P}(A) = \mathrm{P}(B)$. Find the value of r and the value of s



Question 16. Events A and B are such that $\mathrm{P}(A \cap B) = \frac{1}{10}$, $\mathrm{P}(A) = v$ and $\mathrm{P}(B) = w$. It is also given that $\mathrm{P}(A \cup B) = 4\mathrm{P}(A)$ and that $\mathrm{P}((A \cup B)') = \frac{1}{5}$.

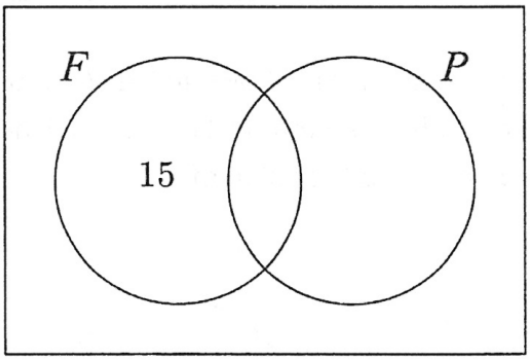
a) Find the value of v and the value of w.

b) Draw a fully labelled Venn diagram showing the events A and B.

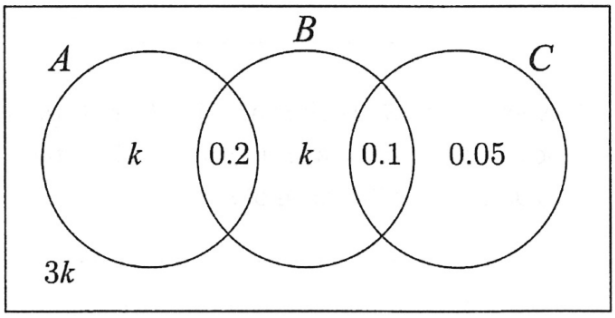
Question 17. 150 adults were surveyed and they were asked which, if either, of the following takeaways they have eaten in the last month: Fish and Chips (F), and Pizza (P). The results are recorded in the table below:

F	P	$F \cap P$
67	108	52

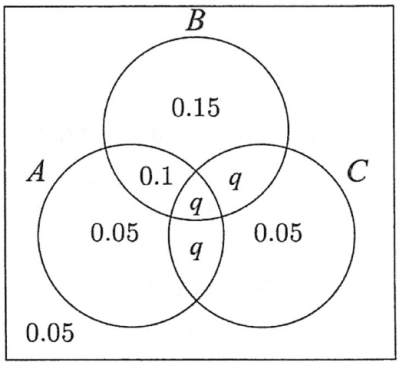
Copy and complete the following Venn diagram to display this information.



Question 18. Events A, B and C are shown in the Venn diagram below. Find the value of k



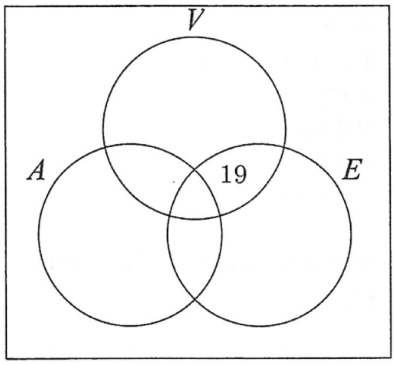
Question 19. Events A, B and C are shown in the Venn diagram below. Find the value of q



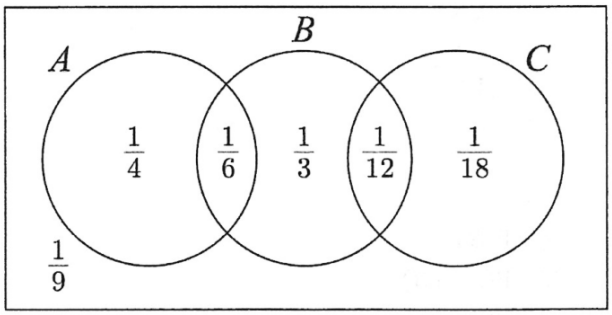
Question 20. 580 adults were surveyed and they were asked which, if any, of the following non-religious holidays they celebrated: April Fools' Day (A), Valentine's Day (V), and Earth Day (E). The results are recorded in the table below:

A	V	E	$A \cap V$	$A \cap E$	$V \cap E$	$A \cap V \cap E$
80	410	30	63	5	23	4

Copy and complete the following Venn diagram to display this information.



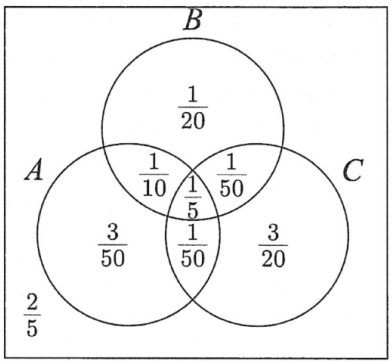
Question 21. Events A, B and C are shown in the Venn diagram below.



Find

a) $\mathrm{P}(B)$
b) $\mathrm{P}(A \cup B)$
c) $\mathrm{P}(B \cup C)$
d) $\mathrm{P}(A \cap B)$
e) $\mathrm{P}(A \cap C)$
f) $\mathrm{P}(A \cup B \cup C)$
g) $\mathrm{P}(C')$
h) $\mathrm{P}(A \cap C')$
i) $\mathrm{P}(B \cap C')$
j) $\mathrm{P}((A \cup B) \cap C)$

Question 22. Events A, B and C are shown in the Venn diagram below.



Find

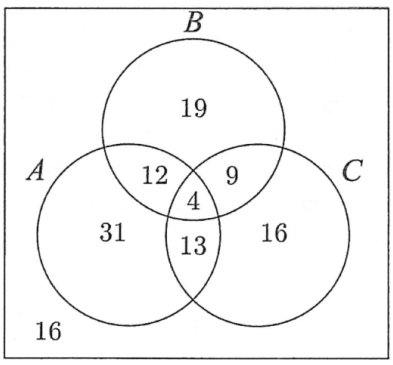
a) $\mathrm{P}(A)$
b) $\mathrm{P}(A \cap B)$
c) $\mathrm{P}(B \cup C)$
d) $\mathrm{P}(A' \cap C)$
e) $\mathrm{P}(A \cap B \cap C)$
f) $\mathrm{P}(A \cup B \cup C)$
g) $\mathrm{P}(A' \cup B')$
h) $\mathrm{P}((A \cup B) \cap C)$
i) $\mathrm{P}((A \cap B) \cup C)$
j) $\mathrm{P}(A' \cup B' \cup C')$

Question 23. In a sixth form college, there are 252 students in Year 12. 85 study both Physics and Chemistry, while 35 do not study either Physics or Chemistry. There are twice as many students who study Physics but not Chemistry, as there are who study Chemistry and not Physics. A student is randomly selected from Year 12. Find the probability that they study Physics.

Question 24. A company employs 360 people. There are certain areas of the company that require a Level 1 or Level 2 passkey. Some employees have both passkeys, some have just one, and others have neither. 105 employees just have a Level 1 passkey and 145 just have a Level 2 passkey. Four times as many employees do not have either passkey as have both. Two members of staff are picked at random. Find the probability that they both have a Level 2 passkey.

Question 25. A company has 150 employees. One day, they have a 'Health and Wellbeing Day' where each employee much choose at least one activity to take part in. There are three activities to choose from: Yoga, Pottery, and a Hike.
77 choose Pottery,
59 choose Yoga,
68 choose a Hike,
22 choose Pottery and Yoga,
19 choose Yoga and a Hike,
21 choose Pottery and a Hike.
An employee is chosen at random. Find the probability that the employee chose all three activities.

Question 26. At a Primary school, there are three after school clubs that students can optionally attend: Art Club (A) on a Monday, Book Club (B) on a Wednesday, and Creative Writing (C) on a Thursday. There are 120 students at the school, and the Venn diagram below shows the number of students who regularly attend each after school club.



Two students are randomly selected.

a) Find the probability that they both attend Art Club and Book Club but not Creative Writing.
b) Find the probability that the first attends Creative Writing and the second attends all three.
c) Find the probability that the two students both attend the same two clubs.

Question 27. In a survey that can be assumed to be a random sample of people living in a large town, 150 adults were asked whether they owned a dog and whether they were under 40 years old. Let A be the event that they owned a dog, and let B be the event that they were under 40 years old. The results are shown in the two-way table below.

	A	A'	Total
B	20	35	55
B'	40	55	95
Total	60	90	150

Find:

a) $\mathrm{P}(A)$
b) $\mathrm{P}(A \cap B)$
c) $\mathrm{P}(B')$
d) $\mathrm{P}(A \cup B)$
e) $\mathrm{P}(A \cap B')$
f) $\mathrm{P}(A \cup B')$

Question 28. In a survey that can be assumed to be a random sample of people living in a large town, 300 adults were asked whether they were vegetarian and whether they were under 40 years old. Let A be the event that they were vegetarian and let B be the event that they were over 40 years old. The results are shown in the two-way table below.

	A	A'	Total
B	20	155	175
B'	15	110	125
Total	35	265	300

Find:

a) $\mathrm{P}(B)$
b) $\mathrm{P}(A' \cap B)$
c) $\mathrm{P}(A' \cup B)$
d) $\mathrm{P}((A \cup B)')$
e) $\mathrm{P}((A \cap B)')$
f) $\mathrm{P}(A' \cup B')$

Question 29. The employees of a large company were surveyed about how they get to and from work. Let A be the event that they travel to and from work by bicycle and let B be the event that they had worked at the company for more than 5 years. The probabilities of these events are shown in the two-way table below.

	A	A'
B	$\frac{2}{19}$	$\frac{7}{19}$
B'	$\frac{1}{38}$	$\frac{1}{2}$

a) Write down fully what is meant by the following events:

i) $A \cap B'$

ii) $A' \cap B$

iii) $A' \cup B$

b) Find:

i) $\mathrm{P}(A)$

ii) $\mathrm{P}(B)$

iii) $\mathrm{P}(A \cap B)$

iv) $\mathrm{P}(A \cup B)$

v) $\mathrm{P}(A' \cap B')$

vi) $\mathrm{P}(A' \cup B')$

Question 30. A doughnut stall offers doughnuts that are dipped in white chocolate, in milk chocolate, both or neither. Let A be the event that the customer orders white chocolate and let B be the event that the customer orders milk chocolate. The percentage of customers ordering each of these are shown in the two-way table below.

	A	A'
B	23%	36%
B'	14%	27%

Find

a) $\mathrm{P}((A' \cup B)')$

b) $\mathrm{P}((A \cap B')')$

Question 31. The two-way table below shows the events A and B. Copy and complete the table, given that $\mathrm{P}(A) = 3\mathrm{P}(B)$.

	A	A'
B	0.19	
B'	0.47	

Question 32. Students at a university must choose one of three science courses: Biology, Chemistry or Physics, and they can opt to complete an additional Maths course if they choose. Let A be the event they choose Biology, let B be the event they choose Chemistry, let C be the event they choose Physics, and let D be the event they choose to study the additional Maths course. The two-way table below shows the percentages that choose each option.

	A	B	C
D	0.1	0.25	0.3
D'	0.2	0.1	0.05

Find

a) $\mathrm{P}(C \cap D')$
b) $\mathrm{P}(B \cup D)$
c) $\mathrm{P}(A \cup B)$
d) $\mathrm{P}((A \cup C) \cap D')$
e) $\mathrm{P}((A \cap D) \cup B)$

Question 33. A supermarket buys onions from two farmers to sell on their shelves. 40% of the onions come from farm A and the rest come from farm B. A certain percentage from each of the farms is spoilt in the transportation process. Overall, 2.1% of the onions are spoilt. The table below collects this information together.

	A	B
supplied	40%	
spoilt	1.5%	

Determine the percentage of onions bought from farm B that is spoilt.

Question 34. A school trip is offered to three classes of students. 25% of the students on the trip come from class A and 45% come from class B. Of the students from class A who are on the trip, 85% are right-handed, and of the students from class B who are on the trip, 70% are right-handed. It is given that 81.25% of the students overall on the trip are right-handed. The table below collects this information together.

	Class A	Class B	Class C
Students on the trip	25%	45%	
Right-handed	85%	70%	

Determine the percentage of students from class C who are on the trip who are right-handed.

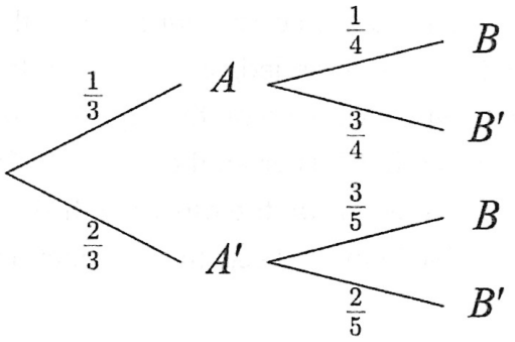
Question 35. The two-way table below shows the events A and B and their respective probabilities. It is given that $3\mathrm{P}(A) = 5\mathrm{P}(B)$. Find the value of p and the value of q

	A	A'
B	p	$3p$
B'	q	$5q$

Question 36. The two-way table below shows the events A, B, C and D and their respective probabilities. It is given that the probability of event B occurring is 26%, and the probability of event A not occurring is 44%. Find the value of r, the value of s and the value of t

	A	A'
B	$2r$	t
C	$3s$	$3r$
D	$2t$	s

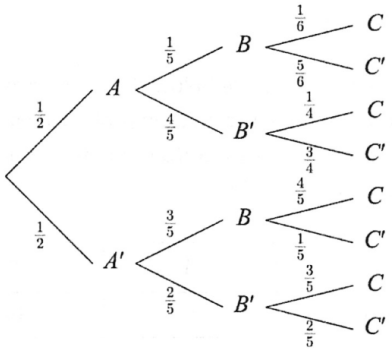
Question 37. The probability tree below shows the events A and B and their respective probabilities.



Find

a) P(A)
b) P(A')
c) P(B)
d) P(B')

Question 38. The probability tree below shows the events A, B and C and their respective probabilities.



Find

a) P(A then B then C)
b) P(A' then B then C')
c) P(C)

Question 39. In a pack of ten playing cards, 8 are blue and 2 are red. One card is randomly chosen from the pack and then replaced. Another card is then randomly chosen from the pack and then replaced.

a) Draw a probability tree to display this information.

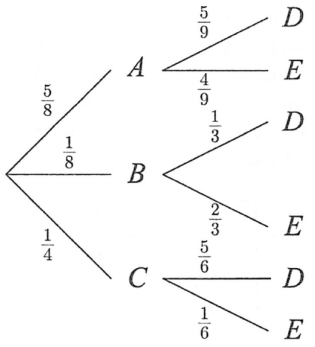
b) Find the probability that the two cards that are chosen are the same colour.

Question 40. In a pack of ten playing cards, 8 are blue and 2 are red. One card is randomly chosen from the pack and is not replaced. Another card is then randomly chosen from the pack and not replaced.

a) Draw a probability tree to display this information.

b) Find the probability that the two cards that are chosen are the same colour.

Question 41. The probability tree below shows the events A, B, C, D and E and their respective probabilities.



Find

a) P(B then D)
b) P(E)
c) P(A or E or both)

Question 42. Daniel usually walks to work and the probability of him arriving before 8:30am is 0.95. If it rains, Daniel catches the bus and then the probability of him arriving before 8:30am is 0.75. The probability on any given day of it raining is 0.2

a) Draw a probability tree to display this information.
b) Find the probability that, on any randomly chosen day, Daniel arrives at work after 8:30am.
c) Five days throughout the year are randomly selected. Find the probability that Daniel arrives at work before 8:30am on all five of these days.

Question 43. In the UK in 2022, 17.47% of the population was aged 14 or under, 63.36% was between the ages of 15 to 64 inclusive, and 19.17% were aged 65 or over. Two people are picked at random from the population.

a) Find the probability that both of them are aged 65 or over.

b) Find the probability that one is aged 65 or over, and the other is aged 14 or under.

Question 44. In a village in the UK in 2022, 46 of the population was aged 14 or under, 194 was between the ages of 15 to 64 inclusive, and 140 were aged 65 or over. Two people are picked at random from the population.

a) Find the probability that both of them are aged 65 or over.

b) Find the probability that one is aged 65 or over, and the other is aged 14 or under.

Question 45. Aisha has six cards numbered 1 to 6 lying face-down on a table. She randomly selects each card and turns it over.

a) Find the probability that she picks the cards in perfect ascending order.
b) Find the probability that the last two cards to be turned over are 5 and then 6
c) Find the probability that the first card to be turned over is 1 and the last is 6
d) Find the probability that the second card to be turned over is 2, the fourth card to be turned over is 4 and the sixth card to be turned over is 6

Question 46. Haroon has eight cards. Five of them are blue and three of them are red, and they are lying face-down on a table. He randomly selects each card and turns it over.

a) Find the probability that he picks all the red cards first, followed by all the blue cards.
b) Find the probability that he picks all the blue cards first, followed by all the red cards.
c) Find the probability that he picks a blue card first and then the colours alternate until there are no more red cards.

Question 47. Pete and Harvey have ten cards numbered 1 to 10 lying face-down on a table. They decide to take it in turns to randomly select cards and turn them over. If someone selects a multiple of 4, then that person automatically wins. If someone selects a prime number, then that person automatically loses. Pete picks first.

a) Find the probability that Pete loses on his first card.
b) Find the probability that Harvey wins on his first card.
c) Find the probability that the maximum number of cards are drawn before someone wins or loses.
d) Find the probability that Pete wins after either of the first two cards being drawn.
e) Find the probability that Pete wins.
f) Find the probability that Harvey wins.

Question 48. Bag A contains 3 red counters and 1 blue counter. Bag B is initially empty. One by one, the counters in Bag A are randomly chosen and placed into Bag B. Find the probability that the first and last counters placed into Bag B are both red.

Question 49. Bag A contains 2 red counters and 1 blue counter. Bag B contains 1 red counter and 3 blue counters. One by one, the counters in Bag A are randomly chosen and placed into Bag B. Find the probability that there are always fewer red counters than blue counters in Bag B.

Question 50. Bag A contains 3 red counters, 10 blue counters and 5 yellow counters. Bag B contains 4 red counters and 5 blue counters. One counter is drawn from Bag A and one counter is drawn from Bag B.

a) Find the probability that both counters are the same colour.

The counters are placed back in their original bags. A counter is now drawn at random from Bag B and placed into Bag A. One counter is then drawn from Bag A and one counter is drawn from Bag B.

b) Find the probability that both counters are the same colour.

Question 51. Bag A contains 8 red counters, 7 yellow counters and 3 green counters. Bag B contains 3 red counters, 2 yellow counters and k green counters. One counter is drawn from Bag A and one counter is drawn from Bag B. The probability of the two colours being different is $\frac{7}{9}$. Find the value of k

Question 52. Bag A contains 4 red counters, 5 yellow counters and 3 green counters. Bag B contains 3 red counters, 2 yellow counters and 1 green counter. One counter is drawn at random from Bag B and placed into Bag A. Then a counter is drawn at random from Bag A and placed into Bag B. Find the probability that Bag B still contains 3 red counters, 2 yellow counters and 1 green counter.

Question 53. Bag A contains 6 red counters, 3 yellow counters and 2 green counters. Bag B contains 5 red counters, 4 yellow counters and 2 green counters. One counter is drawn at random from Bag B and placed into Bag A. Then a counter is drawn at random from Bag A and placed into Bag B. Find the probability that Bag B now has an equal number of red and yellow counters.

Question 54. Janice and Eric are playing with marbles. There are 10 blue marbles and 10 red marbles in a bag. They take it in turns to pick a marble from the bag. If Janice picks a blue marble, or if Eric picks a red marble, the game continues. If Janice picks a red marble, or if Eric picks a blue marble, the game is over. Janice plays first.

a) Draw a probability tree to show the possible results from the first three picks.

b) Find the probability that six marbles are picked from the bag and the game is not over.

Question 55. A sixth form college has 48 students who study A-Level Further Maths across 3 classes. Each student is given the option of teaching a short revision lesson to their peers on one of three topics. The table below shows the number of students from each class who chose each topic.

	Class A	Class B	Class C	Total
Complex Loci	5	6	9	20
Matrices	5	7	6	18
Series	5	3	2	10
Total	15	16	17	48

Three students were randomly selected from the 48 students.
Find the probability that:

a) one taught a lesson on 'Complex Loci', one taught a lesson on 'Matrices', and one taught a lesson on 'Series',
b) all three taught a lesson on 'Complex Loci',
c) one student from each class is picked,
d) two taught a lesson on 'Matrices' and one taught a lesson on 'Series'.

Question 56. Erin and Molly are playing a game where in each round they must complete a puzzle as quickly as possible. The play four rounds, and they decide that someone can only win the game if someone wins two rounds in a row. The probability of Erin winning a round is 0.75 and the probability of Molly winning a round is 0.25

a) Find the probability that Erin wins the game.
b) Find the probability that neither of them wins the game.

Question 57. Cecil and Hilda are to play a game of netball and to practice they make up a short game. They will both get three chances of trying to get the ball in the basket. They must get at least two balls into the basket to win, and a draw is considered a loss for both players. The probability of Cecil scoring is 0.35 and the probability of Hilda scoring is 0.55

a) Find the probability that Cecil wins the game.

b) Find the probability that neither wins the game.